Analysis report for sample: 14ffdcecdd3c6056460cc622fef3b061

This report has been generated by a set of tools for semi-supervised behavioral analysis of executables.

More information about these reports at **http://www.vallejo.cc**.

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Report classification: 2650 (indicates the level of interest of the report, based on the amount of information collected about the sample).

Index

- Sample
- AV detections
- Original source url
- Virustotal url
- Yara matches
- Snapshot
- Threads tree
- Running processes
- Most interesting actions
- Most interesting strings
- Involved hosts
- Dns queries
- Connections and readable network traffic
- Threads behavioural information
- Malware processes' network events
- Sample's unpacked / injected modules

Sample

•md5: 14ffdcecdd3c6056460cc622fef3b061

AV detections

- Microsoft:
- Kaspersky: UDS:DangerousObject.Multi.Generic
- Symantec: Trojan.Trickybot
- Malwarebytes:

Original source url

- hxxp://kissdesign.ca/GktpotdC7dyTH1aoroa.png

Virustotal url

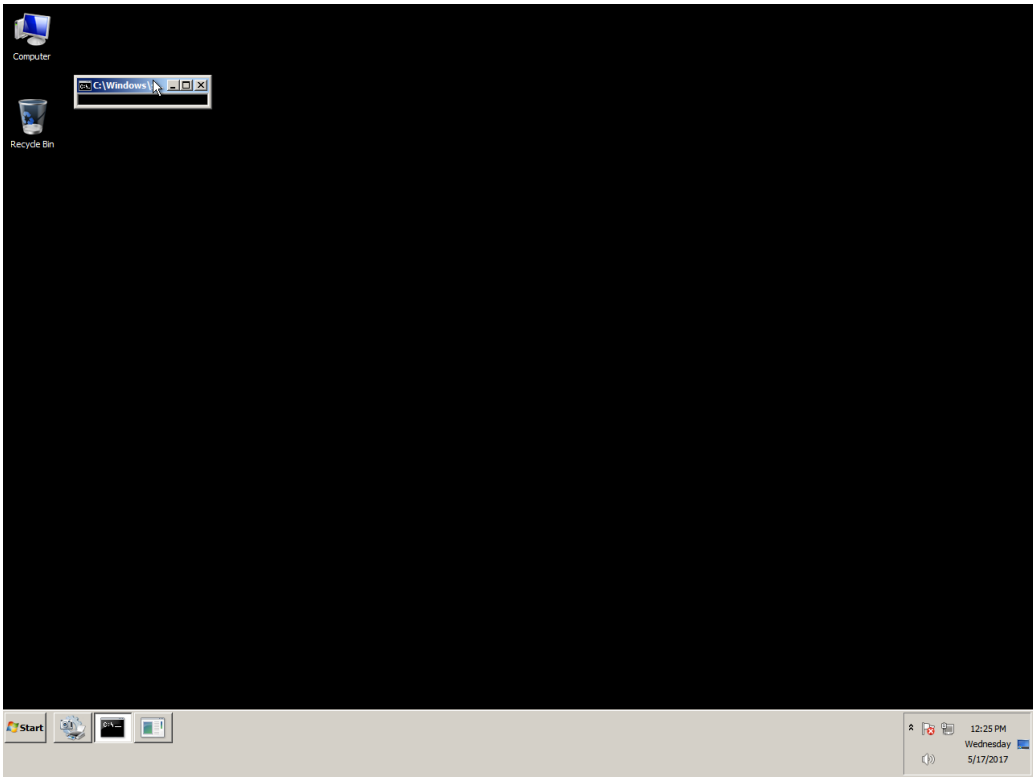
- https://virustotal.com/es/file/be8c214422ab283871308e1f5d6717b967c94399610a4a8f0639511a7a417132/analysis

Yara matches

The following yara rules have matched injected or unpacked modules's code or data areas.

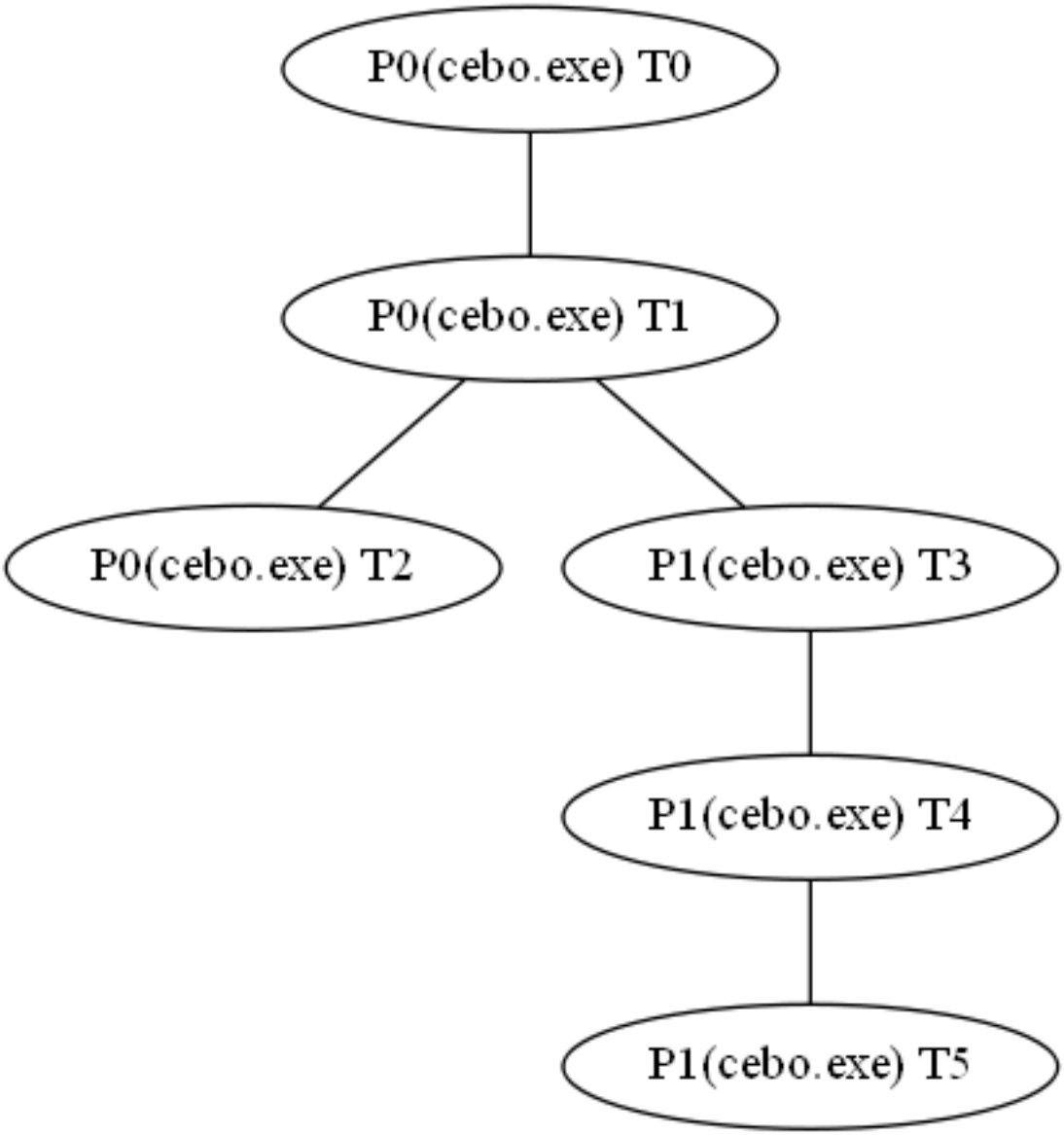
- Armadillo_v171 ()
- Microsoft_Visual_Cpp_v60 ()
- Microsoft_Visual_Cpp_v50v60_MFC_additional ()
- Microsoft_Visual_Cpp_50 ()
- Microsoft_Visual_Cpp_v50v60_MFC ()
- Install_Shield_2000 (Microsoft Visual C++ 5.0)
- Armadillo_v171_additional ()
- Armadillo_v4x ()
- Microsoft_Visual_Cpp ()
- InstallShield_2000_additional ()
- IsPE32 ()
- IsWindowsGUI ()
- HasRichSignature (Rich Signature Check)
- Install_Shield_2000_ (Microsoft Visual C++ 5.0)
- screenshot (Take screenshot)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- HasOverlay (Overlay Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- Advapi_Hash_API (Looks for advapi API functions)

Snapshot


Computer
C:\Windows
Recycle Bin
Start
12:25 PM
Wednesday
5/17/2017

Threads tree

The following tree represents sample's threads. T<index> is an alias for sample's threads (numeration is done in the order of threads creation). P<index> is an alias for processes owning sample's threads.



Running processes

Here it is the list of running processes after sample execution.

Processes with sample' threads

- Name cebo.exe Alias P0
- Name cebo.exe Alias P1

Other processes

- Name System
- Name smss.exe
- Name csrss.exe
- Name wininit.exe
- Name csrss.exe
- Name winlogon.exe
- Name services.exe
- Name lsass.exe
- Name lsm.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name audiodg.exe
- Name TrustedInstaller.exe
- Name svchost.exe
- Name svchost.exe
- Name spoolsv.exe
- Name svchost.exe
- Name svchost.exe
- Name vmtoolse.exe
- Name svchost.exe
- Name dllhost.exe
- Name msdtc.exe
- Name sppsvc.exe
- Name taskhost.exe
- Name dwm.exe
- Name explorer.exe
- Name vmtoolse.exe
- Name SearchIndexer.exe
- Name svchost.exe
- Name wmpnetwk.exe
- Name WmiPrvSE.exe
- Name chrome.exe
- Name WmiPrvSE.exe
- Name SearchFilterHost.exe
- Name taskhost.exe
- Name SearchProtocolHost.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name UI0Detect.exe
- Name pxxx.exe
- Name dllhost.exe

- Name cmd.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name int3.exe
- Name explorer.exe
- Name explorer.exe
- Name taskeng.exe
- Name cebo.exe
- Name cebo.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name cmd.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name explorer.exe
- Name explorer.exe
- Name cmd.exe
- Name conhost.exe
- Name taskkill.exe
- Name mode.com
- Name conhost.exe
- Name WmiPrvSE.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name cmd.exe
- Name conhost.exe
- Name int3.exe

Most interesting actions

The following list it's a collection of the most interesting events captured. This list is ordered by the score assigned to the event. In the section "Threads behavioural information" it's possible to find all the actions performed by each sample's thread ordered chronologically.

- Process Create (C:\\Users\\javi\\AppData\\Roaming\\cebo.exe PID: P1, Command line: C:\\Users\\javi\\AppData\\Roaming\\cebo.exe)

Most interesting strings

The following list it's a collection of the most interesting strings found in the sample's modules code or data. A complete list of strings for each sample's module can be found in the section "Sample's unpacked / injected modules."

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange

- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW

- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId

Involved hosts

- 107.22.214.64:80 (checkip.amazonaws.com)
- 192.168.133.174:49194
- 192.168.133.174:62151
- 192.168.133.1:5355

Dns queries

- 2.133.168.192.in-addr.arpa ---> no answers
- checkip.amazonaws.com ---> 107.22.214.64, 50.19.227.215, 23.21.70.163, 50.19.97.123
- 64.214.22.107.in-addr.arpa ---> no answers
- 158.253.231.177.in-addr.arpa ---> no answers
- dns.msftncsi.com ---> 131.107.255.255

Connections and readable network traffic

The analysis environment captures network traffic happened while sample execution. It classifies the traffic and tries to extract the most significant parts from the communications data.

In this section it's possible to find parts of the communication data that are easier to be read.

• tcp 192.168.133.174:49194 ---> 107.22.214.64 (checkip.amazonaws.com) :80

GET / HTTP/1.1[...]Connection: Keep-Alive[...]User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36[...]Host: checkip.amazonaws.com

• tcp 107.22.214.64 (checkip.amazonaws.com) :80 ---> 192.168.133.174:49194

HTTP/1.1 200 OK[...]Date: Wed, 17 May 2017 19:24:02 GMT[...]Server: lighttpd/1.4.41[...]Content-Length: 14[...]Connection: keep-alive[...]46.25.110.246[...]HTTP/1.1 200 OK[...]Date: Wed, 17 May 2017 19:24:02 GMT[...]Server: lighttpd/1.4.41[...]Content-Length: 14[...]Connection: keep-alive[...]46.25.110.246

• udp 192.168.133.1:5355 ---> 192.168.133.174:62151

in-addr[...]in-addr

Threads behavioural information

In this section it's possible to find information about sample's threads, such as the actions performed by each sample's thread ordered chronologically.

- **Thread T0 (in process P0, cebo.exe) description**
  - **Thread's childs**
    - Thread T1 (in process P0, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T1)
- **Thread T1 (in process P0, cebo.exe) description**
  - **Thread's childs**
    - Thread T2 (in process P0, cebo.exe)
    - Thread T3 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T2)
    - Process Create (C:\\Users\\javi\\AppData\\Roaming\\cebo.exe PID: P1, Command line: C:\\Users\\javi\\AppData\\Roaming\\cebo.exe)
- **Thread T2 (in process P0, cebo.exe) description**
  - **Thread's childs**
  - **Thread' events**
    - Thread Create (Thread ID: TUNKALIAS)
- **Thread T3 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T4 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T4)
- **Thread T4 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T5 (in process P1, cebo.exe)
  - **Thread' events**

- Thread Create (Thread ID: T5)

• Thread T5 (in process P1, cebo.exe) description

• Thread's childs

• Thread' events

- Thread Create (Thread ID: TUNKALIAS)

Malware processes' network events

The analysis environment tries to capture and collect network actions performed by sample's threads.

Sample's unpacked / injected modules

In this section it's possible to find information about sample's modules, such as the rich signatures, strings, yara rules matching the module,...

The analysis environment performs an heuristic search trying to find sample's injected or unpacked modules in memory, dumping and analyzing them.

• Module 1 (probably unpacked / injected by the sample)

• Module 1 rich signatures

• 44616e5300000000000000000000000097895000900000009789300040000000978830045000000c30f6000010000006f76ab00040000000097884001100
0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 1 yara matches

• Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
• Str_Win32_Http_API (Match Windows Http API call)
• IsPE32 ()
• IsWindowsGUI ()
• HasOverlay (Overlay Check)
• HasRichSignature (Rich Signature Check)
• DebuggerCheck__QueryInfo ()
• inject_thread (Code injection with CreateRemoteThread in a remote process)
• network_dns (Communications use DNS)
• win_mutex (Create or check mutex)
• win_token (Affect system token)
• win_files_operation (Affect private profile)
• without_attachments (Rule to detect the no presence of any attachment)
• without_images (Rule to detect the no presence of any image)
• without_urls (Rule to detect the no presence of any url)
• contentis_base64 (This rule finds for base64 strings)
• Advapi_Hash_API (Looks for advapi API functions)

• Module 1 strings

• Module 1 most interesting strings

• url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
• url | valid domains | tlds: S.va>-v
• url | valid domains | tlds: zen.spamhaus.org
• url | valid domains | tlds: cbl.abuseat.org
• url | valid domains | tlds: b.barracudacentral.org
• url | valid domains | tlds: dnsbl-1.uceprotect.net
• url | valid domains | tlds: spam.dnsbl.sorbs.net
• url | valid domains | tlds: checkip.amazonaws.com
• url | valid domains | tlds: ipecho.net
• url | valid domains | tlds: ipinfo.io
• url | valid domains | tlds: api.ipify.org
• url | valid domains | tlds: icanhazip.com
• url | valid domains | tlds: myexternalip.com
• url | valid domains | tlds: wtfismyip.com
• url | valid domains | tlds: ip.anysrc.net
• url | valid domains | tlds: spam.dnsbl.sorbs.net
• url | valid domains | tlds: dnsbl-1.uceprotect.net
• url | valid domains | tlds: b.barracudacentral.org

- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo

- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA

- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 1 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3

- 2(2,20282P2T2I2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2??2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2

- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts

- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control

- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>

- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- >Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count

- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- >Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\

- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QWWh09=
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<

- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;:;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- >%s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\

- %s%s_configs\\
- Content-Length: %d
- >D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*>
- v%PQh@*>
- WQVRPh
- SQRPh|)>
- PQRhT)>
- SWPQRh8)>
- VVhp->
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- QRhx3>
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA>
- QPh`A>
- PSSSSSSSj
- jjjjjj

• Module 2 (probably unpacked / injected by the sample)

• Module 2 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000 00c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 2 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)

- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 2 strings

• Module 2 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net

- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData

- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW

- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 2 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2?2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4

- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7

- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z

- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>

- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- >Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT

- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain

- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- >Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QWWh09=
- @.reloc
- @.data
- `.rdata

- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;:;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v

- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- >%s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- >D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*>
- v%PQh@*>
- WQVRPh
- SQRPh|)>
- PQRhT)>
- SWPQRh8)>
- VVhp->
- D$$_^[
- |$,run
- |$(ver

- |$(mcco
- QRhx3>
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA>
- QPh`A>
- PSSSSSSSj
- jjjjjj

• Module 3 (probably unpacked / injected by the sample)

• Module 3 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000000c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 3 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 3 strings

• Module 3 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com

- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource

- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait

- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 3 other strings

• 0 080<0@0
• 4 4(4<4D4X4`4h4p4x4
• 3(3,3<3@3D3L3d3t3x3
• 2(2,20282P2T2l2|2
• $2(2,2024282<2@2
• $30343`3d3h3l3p3t6x6|647(<
• 687>7D7J7P7V7]7d7k7r7y7
• 6%6/6@6T6\\6}6
• 5A5L5R5
• 40464=4Q4V4
• 2)3.34393>3C3H3N3V3^3u3
• 0!0-1H2P2h2
• 455;5c5u5
• 2"3L3S3
• 2!2+2_2l2
• >e?o?|?
• 8(8;8E8g8t8
• 7;7E7m7r7
• 4S5]5c5i5p5x5
• 2&2/2??2l2
• >,>2>7>A>
• 828O8_8e8k8r8z8
• 7 7'7-747E7
• 292A2K2
• ;3;O;k;t;
• 9#9:9A9F9X9]9b9n9s9x9
• 7/797|7
• 696B6H6N6U6]6j6r6
• 2E3e4~4
• 0Z091Z1
• 9$9=9D9X9o9
• 8I8W8o8y8
• 737:7O7V7
• 3H475L5
• 1E1U1n1~1
• $0.080
• 9E:O:\\:f:s:|:
• 8.888O8
• 5+545A5K5~5
• 4"444E4V4_4
• 363>3N3X3
• 2"2*282=2H2M2
• 0(0-0:0T0
• 8)9C9w9
• 5E6O6U6[6b6j6w6
• 5$5+535A5I5W5_5m5u5
• 3<3G3Y3`3
• 2 2(22272?2H2M2U2_2d2l2v2{2
• 1(1W1]1c1i1o1}1
• 0 070C0N0U0f0o0t0}0
• ?_?d?i?o?u?
• 7J7[7o7
• 6)636?6I6
• 5/585M5W5a5k5

- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable

- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken

- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>

- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error

- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart

- Modules\\
- Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QRhx3
- QWWh09
- QPh`A
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>

- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;:;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- %s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/

- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*
- v%PQh@*
- WQVRPh
- SQRPh|)
- PQRhT)
- SWPQRh8)
- VVhp-
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA
- PSSSSSSSj
- jjjjjj

• Module 4 (probably unpacked / injected by the sample)

• Module 4 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000 00c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 4 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()

- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 4 strings

• Module 4 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net

- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower

- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW

- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 4 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2?2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7

- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5

- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit

- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>

- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf

- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2

- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4

- |$(servu<f
- |$(gtagug
- QRhx3
- QWWh09
- QPh`A
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;:;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--

- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- %s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*
- v%PQh@*

- WQVRPh
- SQRPh|)
- PQRhT)
- SWPQRh8)
- VVhp-
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA
- PSSSSSSSj
- jjjjjjj

• Module 5 (probably unpacked / injected by the sample)

• Module 5 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000000c30f6000010000006f76ab00040000000097884001100000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 5 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 5 strings

• Module 5 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net

- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory

- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread

- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW

- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 5 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2?2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1

- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
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- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll

- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject

- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>

- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished

- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control

- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QRhx3
- QWWh09
- QPh`A
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;

- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- :!;+;4;::;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- %s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/

- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*
- v%PQh@*
- WQVRPh
- SQRPh|)
- PQRhT)
- SWPQRh8)
- VVhp-
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA
- PSSSSSSSj
- jjjjjjj

• Module 6 (probably unpacked / injected by the sample)

• Module 6 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000000c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 6 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 6 strings

• Module 6 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org

- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange

- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW

- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 6 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7

- 4S5]5c5i5p5x5
- 2&2/2?2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6

- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode

- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>

- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- Author Name
- services update

- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7

- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3

- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QRhx3
- QWWh09
- QPh`A
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;::;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln

- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- %s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj

- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*
- v%PQh@*
- WQVRPh
- SQRPh|)
- PQRhT)
- SWPQRh8)
- VVhp-
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA
- PSSSSSSSj
- jjjjjj

• Module 7 (probably unpacked / injected by the sample)

• Module 7 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000 00c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 7 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 7 strings

• Module 7 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v

- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread

- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW

- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW
- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc

- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 7 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2
- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2?2I2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9

- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7
- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile

- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp
- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll

- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>
- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>

- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>
- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- !Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id

- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown
- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed

- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- !Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip
- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QWWh09
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;

- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- ;!;+;4;::;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll
- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s

- !%s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- !D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t
- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*!
- v%PQh@*!
- WQVRPh
- SQRPh|)!
- PQRhT)!
- SWPQRh8)!
- VVhp-!
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- QRhx3!
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA!
- QPh`A!
- PSSSSSSSj
- jjjjjj

• Module 8 (probably unpacked / injected by the sample)

• Module 8 rich signatures

- 44616e530000000000000000000000000978950009000000097893000400000009788300450000 00c30f6000010000006f76ab00040000000097884001100 0000c30f5f0001000000c30f5d001700000000000100e800000027c67e00010000001b9daf00160000001b9d9a00010000001b9d9d00

• Module 8 yara matches

- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_dns (Communications use DNS)
- win_mutex (Create or check mutex)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Advapi_Hash_API (Looks for advapi API functions)

• Module 8 strings

• Module 8 most interesting strings

- url | http tag | valid domain | tlds: xmlns="http://schemas.microsoft.com/windows/2004/02/mit/task">
- url | valid domains | tlds: S.va>-v
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: spam.dnsbl.sorbs.net
- url | valid domains | tlds: dnsbl-1.uceprotect.net
- url | valid domains | tlds: b.barracudacentral.org
- url | valid domains | tlds: cbl.abuseat.org
- url | valid domains | tlds: zen.spamhaus.org
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com

- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- url | valid domains | tlds: checkip.amazonaws.com
- url | valid domains | tlds: ipecho.net
- url | valid domains | tlds: ipinfo.io
- url | valid domains | tlds: api.ipify.org
- url | valid domains | tlds: icanhazip.com
- url | valid domains | tlds: myexternalip.com
- url | valid domains | tlds: wtfismyip.com
- url | valid domains | tlds: ip.anysrc.net
- api | frequently used by malware: QueryPerformanceCounter
- api | frequently used by malware: GetAdaptersInfo
- api | frequently used by malware: NtQueryInformationProcess
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: CryptGetHashParam
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: VirtualProtectEx
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: LoadResource
- api | frequently used by malware: CreateThread
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: WriteFile
- api | frequently used by malware: GetVersion
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: ExitProcess
- api | ws2_32: getaddrinfo
- api | ws2_32: freeaddrinfo
- api | crypt32: CryptBinaryToStringW
- api | crypt32: CryptStringToBinaryW
- api | wininet: InternetCanonicalizeUrlW
- api | kernel32: GetSystemTimeAsFileTime
- api | kernel32: GetCurrentProcessId

- api | kernel32: GetCurrentThreadId
- api | kernel32: GetModuleHandleA
- api | kernel32: SetUnhandledExceptionFilter
- api | kernel32: UnhandledExceptionFilter
- api | kernel32: InterlockedCompareExchange
- api | kernel32: InterlockedExchange
- api | kernel32: RtlUnwind
- api | ntdll: memcpy
- api | ntdll: memset
- api | ntdll: _vsnprintf
- api | ntdll: tolower
- api | ntdll: _vsnwprintf
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | advapi32: DuplicateTokenEx
- api | kernel32: CreateProcessAsUserW
- api | advapi32: EqualSid
- api | advapi32: CryptHashData
- api | advapi32: CryptDestroyHash
- api | advapi32: CryptDecrypt
- api | advapi32: CryptDestroyKey
- api | advapi32: CryptCreateHash
- api | advapi32: CryptImportKey
- api | advapi32: CryptReleaseContext
- api | advapi32: CryptSetKeyParam
- api | advapi32: CryptAcquireContextW
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: LookupAccountSidW
- api | advapi32: GetTokenInformation
- api | advapi32: GetUserNameW
- api | kernel32: LocalFree
- api | kernel32: CreateMutexW
- api | kernel32: lstrcpyW
- api | kernel32: DuplicateHandle
- api | kernel32: Process32NextW
- api | kernel32: CreateEventW
- api | kernel32: GetExitCodeThread
- api | kernel32: Process32FirstW
- api | kernel32: ResetEvent
- api | kernel32: TerminateProcess
- api | kernel32: VirtualFreeEx
- api | kernel32: SetEvent
- api | kernel32: SignalObjectAndWait
- api | kernel32: WaitForSingleObject
- api | kernel32: CreateDirectoryW
- api | kernel32: lstrlenA
- api | kernel32: lstrcmpA
- api | kernel32: LoadLibraryA
- api | kernel32: GetFileTime
- api | kernel32: FindNextFileW
- api | kernel32: GetSystemInfo
- api | kernel32: LockResource
- api | kernel32: FindClose
- api | kernel32: GetLastError
- api | kernel32: MultiByteToWideChar
- api | kernel32: lstrcpynW
- api | kernel32: LoadLibraryW

- api | kernel32: GetModuleHandleW
- api | kernel32: FreeLibrary
- api | kernel32: FindResourceW
- api | kernel32: FindFirstFileW
- api | kernel32: GetFullPathNameW
- api | kernel32: lstrlenW
- api | kernel32: lstrcmpW
- api | kernel32: GetComputerNameW
- api | kernel32: SetFileAttributesW
- api | kernel32: DeleteFileW
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetTempPathW
- api | kernel32: GetStartupInfoW
- api | kernel32: GetModuleFileNameW
- api | kernel32: GetFileAttributesW
- api | kernel32: LeaveCriticalSection
- api | kernel32: CopyFileW
- api | kernel32: InitializeCriticalSectionAndSpinCount
- api | kernel32: MoveFileExW
- api | kernel32: CreateProcessW
- api | kernel32: GetTempFileNameW
- api | kernel32: lstrcmpiW
- api | kernel32: CreateFileW
- api | kernel32: ReadFile
- api | kernel32: SetFilePointer
- api | kernel32: CloseHandle
- api | kernel32: GetVersionExW
- api | kernel32: GetCurrentProcess
- api | kernel32: InterlockedDecrement
- api | kernel32: HeapReAlloc
- api | kernel32: HeapFree
- api | kernel32: GetProcessHeap
- api | kernel32: HeapAlloc
- api | kernel32: WTSGetActiveConsoleSessionId
- api | kernel32: LoadLibraryW
- api | kernel32: ResetEvent
- api | kernel32: CloseHandle
- api | kernel32: WaitForSingleObject
- api | kernel32: SignalObjectAndWait

• Module 8 other strings

- 0 080<0@0
- 4 4(4<4D4X4`4h4p4x4
- 3(3,3<3@3D3L3d3t3x3
- 2(2,20282P2T2l2|2
- $2(2,2024282<2@2
- $30343`3d3h3l3p3t6x6|647(<
- 687>7D7J7P7V7]7d7k7r7y7
- 6%6/6@6T6\\6}6
- 5A5L5R5
- 40464=4Q4V4
- 2)3.34393>3C3H3N3V3^3u3
- 0!0-1H2P2h2

- 455;5c5u5
- 2"3L3S3
- 2!2+2_2l2
- >e?o?|?
- 8(8;8E8g8t8
- 7;7E7m7r7
- 4S5]5c5i5p5x5
- 2&2/2?2l2
- >,>2>7>A>
- 828O8_8e8k8r8z8
- 7 7'7-747E7
- 292A2K2
- ;3;O;k;t;
- 9#9:9A9F9X9]9b9n9s9x9
- 7/797|7
- 696B6H6N6U6]6j6r6
- 2E3e4~4
- 0Z091Z1
- 9$9=9D9X9o9
- 8I8W8o8y8
- 737:7O7V7
- 3H475L5
- 1E1U1n1~1
- $0.080
- 9E:O:\\:f:s:|:
- 8.888O8
- 5+545A5K5~5
- 4"444E4V4_4
- 363>3N3X3
- 2"2*282=2H2M2
- 0(0-0:0T0
- 8)9C9w9
- 5E6O6U6[6b6j6w6
- 5$5+535A5I5W5_5m5u5
- 3<3G3Y3`3
- 2 2(22272?2H2M2U2_2d2l2v2{2
- 1(1W1]1c1i1o1}1
- 0 070C0N0U0f0o0t0}0
- ?_?d?i?o?u?
- 7J7[7o7
- 6)636?6I6
- 5/585M5W5a5k5
- 2-272O2X2e2o2y2
- 8G8P8X8g8m8s8z8
- 6:7D7Q7[7
- 5#6<6}6
- _1i1p1`2m2v2
- 162M2`2g2
- 0#1,12181?1G1P1X1a1i1r1z1
- L0x0|0
- 8-878A8
- 6 6/6z6
- 5#5-575z5
- 1`2j2w2
- 0,0i0p0
- 8&8+898V8
- 7#7J7T7^7k7u7{7

- 6 61666D6m6w6
- 4?5H5Y5g5
- 3&3o3}3
- 1C1J1Z1i1
- 949:9E9K9p9z9
- 6)666@6J6T6]6g6
- 4%4/4<4F4K4U4d4s4
- $0.0Q0Z0g0q0~0
- =%=9=E=J=P=]=b=o=u=
- 526B6L6q6
- 5:5?5E5K5Q5W5
- 2-3#4)474E4S4Y4`4k4
- 1#1V1`1j1
- 0V0g0|0
- >4>U>e>{>
- 8#8,868L8
- 7 707D7V7\\7j7
- 5S6^6t6
- "0]0r0O1b1
- =nD3W>8
- .?AVServerConfig@@
- .?AVModule@@
- .?AVExtServerConfig@@
- .?AVClientConfig@@
- Mar 15 1998
- Mar 15 1998
- .?AVBConfig@@
- .?AV_com_error@@
- .?AVexception@@
- .?AVbad_alloc@std@@
- USERENV.dll
- UnloadUserProfile
- DestroyEnvironmentBlock
- CreateEnvironmentBlock
- LoadUserProfileW
- WS2_32.dll
- iphlpapi.dll
- WinHttpCrackUrl
- WinHttpReadData
- WinHttpOpenRequest
- WinHttpOpen
- WinHttpQueryDataAvailable
- WinHttpQueryHeaders
- WinHttpCloseHandle
- WinHttpConnect
- WinHttpSendRequest
- WinHttpSetOption
- WinHttpSetTimeouts
- WinHttpReceiveResponse
- PathRemoveBackslashW
- PathFindExtensionW
- PathRenameExtensionW
- PathCombineW
- PathAddBackslashW
- PathRemoveFileSpecW
- PathFindFileNameW
- _controlfp

- ?terminate@@YAXXZ
- _onexit
- __dllonexit
- _unlock
- __set_app_type
- __p__fmode
- __p__commode
- __setusermatherr
- _amsg_exit
- _initterm
- _cexit
- __wgetmainargs
- ??1type_info@@UAE@XZ
- malloc
- _callnewh
- _CxxThrowException
- ??0exception@@QAE@ABV0@@Z
- ??1exception@@UAE@XZ
- ?what@exception@@UBEPBDXZ
- ??0exception@@QAE@XZ
- _time64
- _localtime64
- wcsftime
- OLEAUT32.dll
- ole32.dll
- CoInitializeEx
- CoInitializeSecurity
- CoUninitialize
- SHELL32.dll
- SHGetFolderPathW
- ADVAPI32.dll
- KERNEL32.dll
- BCryptCloseAlgorithmProvider
- BCryptVerifySignature
- BCryptImportKeyPair
- BCryptOpenAlgorithmProvider
- NCryptFreeObject
- NCryptDeleteKey
- NCryptImportKey
- NCryptOpenStorageProvider
- kernel32.dll
- WTSQueryUserToken
- WTSFreeMemory
- WTSEnumerateSessionsA
- wtsapi32
- Release
- FreeBuffer
- Control
- <moduleconfig>*</moduleconfig>
- Content-Disposition: form-data; name="%S"
- ------Boundary%08X
- Fbad allocation
- Buo>Bu
- -v99.v`
- a/v+E.v
- CONFIG
- <RunLevel>LeastPrivilege</RunLevel>

- <LogonType>InteractiveToken</LogonType>
- </UserId>
- <LogonType>InteractiveToken</LogonType>
- <GroupId>NT AUTHORITY\\SYSTEM</GroupId>
- <RunLevel>HighestAvailable</RunLevel>
- <Enabled>true</Enabled>
- <LogonTrigger>
- </Actions>
- </Exec>
- </Command>
- <UserId>
- <Command>
- <Exec>
- <Actions Context="Author">
- <Priority>7</Priority>
- <ExecutionTimeLimit>PT0S</ExecutionTimeLimit>
- <WakeToRun>false</WakeToRun>
- <RunOnlyIfIdle>false</RunOnlyIfIdle>
- <Hidden>true</Hidden>
- <Enabled>true</Enabled>
- <AllowStartOnDemand>true</AllowStartOnDemand>
- </IdleSettings>
- <RestartOnIdle>false</RestartOnIdle>
- <StopOnIdleEnd>true</StopOnIdleEnd>
- <IdleSettings>
- <RunOnlyIfNetworkAvailable>false</RunOnlyIfNetworkAvailable>
- <StartWhenAvailable>true</StartWhenAvailable>
- <AllowHardTerminate>false</AllowHardTerminate>
- <StopIfGoingOnBatteries>false</StopIfGoingOnBatteries>
- <DisallowStartIfOnBatteries>false</DisallowStartIfOnBatteries>
- <MultipleInstancesPolicy>IgnoreNew</MultipleInstancesPolicy>
- <Settings>
- </Principals>
- </Principal>
- <Principal id="Author">
- <Principals>
- </CalendarTrigger>
- </ScheduleByDay>
- <DaysInterval>1</DaysInterval>
- <ScheduleByDay>
- <Enabled>true</Enabled>
- </StartBoundary>
- </UserId>
- <StartBoundary>
- </Repetition>
- <StopAtDurationEnd>false</StopAtDurationEnd>
- <Duration>P1D</Duration>
- <Interval>PT3M</Interval>
- <Repetition>
- <CalendarTrigger>
- <UserId>
- <Triggers>
- </RegistrationInfo>
- <URI>\\ServiceTask</URI>
- <Description>Look for services monitor.</Description>
- <Version>1.0.1</Version>
- <RegistrationInfo>

- <Task version="1.2"
- <?xml version="1.0" encoding="UTF-16"?>
- </LogonTrigger>
- Trigger1
- !Author Name
- services update
- group_tag
- group_tag
- CONFIG
- config.conf
- config.conf
- not listed
- listed
- NAT status
- client is not behind NAT
- NAT status
- client is behind NAT
- NAT status
- failed
- Global\\VLock
- /plain
- /plain/clientip
- svchost.exe
- Trigger1
- Author Name
- services update
- module
- SignatureLength
- ECCPUBLICBLOB
- ECDSA_P384
- client_id
- client_id
- Win32 error
- Control failed
- Process has been finished
- Module was unloaded
- Process has been finished
- Module already unloaded
- release
- Start failed
- Process was unloaded
- GetParentInfo error
- Win32 error
- Unable to load module from server
- Win32 error
- Decode from BASE64 error
- Win32 error
- Invalid params count
- No params
- noname
- svchost.exe
- /plain
- /plain/clientip
- plugins
- servconf
- Wininet
- Unknown

- Windows 2000
- Windows XP
- Windows Server 2003
- Windows Vista
- Windows Server 2008
- Windows 7
- Windows Server 2008 R2
- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows 10 Server
- okernel32.dll
- 0.0.0.0
- /plain/clientip
- /plain
- ECCPUBLICBLOB
- Microsoft Software Key Storage Provider
- winsta0\\default
- Modules\\
- kernel32
- kernel32
- skernel32
- kernel32
- ekernel32
- tkernel32
- tkernel32
- Load to M failed
- Run D failed
- Load to P failed
- Find P failed
- Create ZP failed
- Module has already been loaded
- parentfiles
- period
- control
- needinfo
- autocontrol
- autoconf
- processname
- autostart
- Modules\\
- !Module is not valid
- svchost.exe
- /plain
- /plain/clientip
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- Modules\\
- svchost.exe
- /plain
- /plain/clientip

- Content-Type: multipart/form-data; boundary=%s
- ------Boundary%08X
- Global\\VLock
- t19w ~#3
- 9^ ~-W3
- 9^H~%3
- um9F`u
- u79F,t
- 9sTaru
- |$(autou
- |$,su4
- |$(servu<f
- |$(gtagug
- QWWh09
- @.reloc
- @.data
- `.rdata
- !This program cannot be run in DOS mode.
- ;G;j;w;
- :$:*:0:5:R:
- ?/?:?m?
- >#>0>D>
- :H=R=_=l=r=
- >%>+>2>:>G>O>\\>d>q>y>
- 9':-:s:|:
- <!=B=N=T=`=f=m=s=
- =F=S=]=w=|=
- >(?,?0?4?h?q?
- =!=.=8=E=N=\\=f=
- ?$?<?D?\\?d?|?
- >,><>[>
- ;E;N;g;q;
- >X>g>q>w>}>
- <X=g=q=w=}=
- ;#;1;;;D;N;^;l;
- ?$?9?F?L?g?p?
- >T>_>e>k>r>z>
- :':1:A:V:f:l:r:y:
- ?$?<?E?
- >!>*>7>A>N>X>m>v>
- =h=t=z=
- ;-;3;9;@;H;U;];f;n;w;
- >$>?>S>
- :&:0:=:G:U:
- >=?F?Y?
- <)<D<L<]<u<
- >+>3>D>I>T>d>n>{>
- < <,<2<8<?<G<W<_<o<w<
- ;&;/;S;[;h;~;
- ><?W?h?v?
- :!;+;4;::;V;
- PADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDING
- ng<{iF<
- WINHTTP.dll
- ntdll.dll

- SHLWAPI.dll
- StrStrIW
- CRYPT32.dll
- _wcmdln
- _XcptFilter
- msvcrt.dll
- BCryptDestroyKey
- BCryptGetProperty
- --%s--
- xppwpp
- ```hhh
- NKagj(h
- >vH2>vb\\>vq
- J-vM)v
- 8.v&<.v
- ,v1#-v
- -v%9-v
- -v\\9.vcv.v
- ,v(7.v:
- -vM7.v
- -vCE.v
- ,v]7.v
- UQPXY]Y[
- URPQQh
- SYSTEM
- </Task>
- </Settings>
- %Y-%m-%dT%H:%M:%S
- </Triggers>
- SYSTEM
- %s start
- %s.%s.%s.%s
- !%s.%s
- %s_W%d%d%d.
- %s/%s/64/%s/%s/%s/
- %s/%s/63/%s/%s/%s/%s/
- /%s/%s/25/%s/
- /%s/%s/23/%d/
- /%s/%s/14/%s/%s/0/
- /%s/%s/10/%s/%s/%d/
- /%s/%s/5/%s/
- /%s/%s/1/%s/
- /%s/%s/0/%s/%s/%s/%s/%s/
- Bcrypt.dll
- Ncrypt.dll
- %s %s SP%d
- Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/57.0.2987.133 Safari/537.36
- %s%s_configs\\
- %s%s_configs\\
- Content-Length: %d
- !D:(A;;GA;;;WD)(A;;GA;;;BA)(A;;GA;;;SY)(A;;GA;;;RC)
- t)9^`t$
- D:$f;O
- 9wXt VS
- 9_`t<h
- D$$_^[
- t"9~`t

- WSSSS3
- WWWWWWR
- QWWWRh
- QSSSSSSSj
- VVPQVR
- ?sTart
- PVVVVVVVW
- v&SPhL;*!
- v%PQh@*!
- WQVRPh
- SQRPh|)!
- PQRhT)!
- SWPQRh8)!
- VVhp-!
- D$$_^[
- |$,run
- |$(ver
- |$(mcco
- QRhx3!
- WSSSS3
- ;XH}Bj
- t$,SSSSSSV
- t4+D$P
- QxVhpA!
- QPh`A!
- PSSSSSSSj
- jjjjjj